

The Process of Photosynthesis

Photosynthesis is the process by which green plants convert light energy into chemical energy stored in glucose.

The reaction is summarised as $6\text{CO}_2 + 6\text{H}_2\text{O} + \text{light}$ giving $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$. This takes place inside the chloroplasts of plant cells, which contain the green pigment chlorophyll.

Photosynthesis occurs in two stages: the light-dependent reactions, which require sunlight and produce ATP and NADPH, and the light-independent Calvin cycle, which fixes carbon dioxide.

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